

Cancer Risk in Patients With Empyema

A Nationwide Population-Based Study

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Abstract: This study aimed to evaluate cancer risk and possible risk factors in patients diagnosed with empyema.

A total of 31,636 patients with newly diagnosed empyema between January 1, 1999 and December 31, 2010 were included in this study. Standardized incidence ratios (SIRs) were calculated to compare the cancer incidence in these empyema patients to that in the general population. Adjusted hazard ratios were also calculated to investigate whether characteristics increased cancer risk.

During the 12-year study period, 2,654 cancers occurred in 31,636 patients with empyema, yielding an SIR of 2.67 (95% confidence interval [CI] 2.57–2.78). We excluded cancer that occurred within 1 year to avoid surveillance bias. The cancer risk remained significantly increased (SIR 1.50, 95% CI 1.41–1.58). Specifically, patients with empyema had higher SIR of cancers of the head and neck (1.50, 95% CI 1.41–1.58), esophagus (2.56, 95% CI 1.92–3.33), stomach (1.49, 95% CI 1.16–1.89), liver and biliary tract (2.18, 95% CI 1.93–2.45), and lung and mediastinum (1.62, 95% CI 1.39–1.86). Age ≥ 60 , male sex, diabetes mellitus, and liver cirrhosis were independent risk factors for cancer development.

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Our study demonstrates an increased incidence of cancer development in patients with empyema, and patients' age ≥ 60 , men, and those with diabetes mellitus and liver cirrhosis showed a higher incidence of developing cancer compared to the general population. The association between such kind of infection and secondary malignancy may be elucidated by further study.

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Abbreviations: CI = confidence interval, HR = hazard ratio, LHID = Longitudinal Health Insurance Database, NHI = National Health Insurance, NHIRD = National Health Insurance Research Dataset, NHRI = National Health Research Institutes, SIR = standardized incidence ratio.

INTRODUCTION

Empyema is the accumulation of pus in the thoracic cage that often results from a progressive deposition of parapneumonic fluid. Persistent infection eventually results in the formation of scar tissue and a pleural peel encompassing the lung.¹ Even with the advance of antibiotics and pneumococcal vaccines, this still accounts for approximately 5% of cases of pneumonia.² Incidence of empyema has been increasing worldwide in recent decades.³ About 15% of empyema patients die and 30% of patients require interventional drainage of the pleural space.^{4,5} Most studies focus on short-term empyema complications; however, the long-term effects of this disease, including cancer, have hardly been studied.

Increased cancer risk in patients with inflammatory and infectious diseases has been reported in many studies.^{6,7} Furthermore, the effects of chronic pulmonary inflammatory disease on cancer have been well documented.^{8,9} However, no study to date has focused on the association between empyema and further cancer risk.

Using the National Health Insurance Research Dataset (NHIRD) of Taiwan, we conducted a nationwide population-based study to examine the relative risk of malignancies, including specific cancer types, in patients with empyema.

MATERIALS AND METHODS

Data Sources

Taiwan's National Health Insurance (NHI) program, which began in 1995, is a mandatory universal health insurance program that covers more than 99% of the Taiwanese population.¹⁰ The NHIRD is managed by the National Health Research Institutes (NHRI) of Taiwan and consists of detailed healthcare data from more than 28 million enrollees. It includes coverage for outpatient, inpatient, emergency, dental, and traditional Chinese medicine services. The Longitudinal Health Insurance Database is a subset of the NHIRD and is a representative database containing 1,000,000 patients randomly